Spark2 SDK: Repository Structure and API

- Language: C++20

Repository Structure

The pre-built C++ distribution lives under `spark2_cpp_dist/` in the Spark2 SDK repository. Shared robot configuration is bundled at the repository root in `configuration/`.

```
spark2_sdk/
|-- configuration/              (robot configuration)
|-- spark2_cpp_dist/
|   |-- CMakelists.txt
|   |-- spark2_sdk_cpp.md
|   |-- include/
|   |   |-- spark2.h            (main SDK class API)
|   |   |-- configurator.h      (configuration API)
|   |   |-- kinematics.h        (kinematics API)
|   |   `-- types.h             (shared data types/enums)
|   |-- lib/                    (pre-built shared library)
|   `-- examples/
|       |-- move_point.cpp
|       |-- move_path.cpp
|       |-- teach_and_playback.cpp
|       |-- set_configuration.cpp
|       `-- calculate_kinematics.cpp
```

Core SDK API (`Spark2`)

API	Description	Group
`Spark2(config_prefix_path="")`	Create SDK client; optional path prefix for configuration files.	constructor
`start(), stop()`	Start or stop arm runtime session.	connection
`enableArmJoint(arm_state)`	Enable arm joints by 6-element boolean state array.	connection
`setArmSmoothingMethod(method)`	Select smoothing strategy for single-target arm motion.	motion mode
`movePos(arm_pos, v=50, t=0)`	Command joint-space target position.	manual motion
`movePosPath(path, vPath, tPath)`	Command sequence of joint-space waypoints.	manual motion

API	Description	Group
`moveVel(arm_vel, a=10, t=0)`	Command joint-space velocity.	manual motion
`moveEEPoint(pose, v=50, t=0)`	Move end-effector to cartesian pose (point mode).	manual motion
`moveEEPointPath(path, vPath, tPath)`	Move through end-effector pose waypoints (point mode).	manual motion
`moveEELine(pose, v=50, t=0)`	Move end-effector to cartesian pose (line mode).	manual motion
`moveEELinePath(path, vPath, tPath)`	Move through end-effector pose waypoints (line mode).	manual motion
`moveToolPoint(pose, v=50, t=0)`	Move tool frame to cartesian pose (point mode).	manual motion
`moveToolPointPath(path, vPath, tPath)`	Move tool frame through cartesian waypoints (point mode).	manual motion
`moveToolLine(pose, v=50, t=0)`	Move tool frame to cartesian pose (line mode).	manual motion
`moveToolLinePath(path, vPath, tPath)`	Move tool frame through cartesian waypoints (line mode).	manual motion
`goHome(v=50, t=0)`	Move robot arm to predefined home pose.	manual motion
`moveGripperPos(pos, v=50, t=0)`	Command gripper joint position.	manual motion
`startTeach()`, `stopTeach()`	Enter or exit teach mode.	teach mode
`startPlayback()`, `stopPlayback()`, `resetPlayback()`	Control playback mode lifecycle.	playback mode
`getPos()`, `getVel()`, `getTor()`	Read current joint position, velocity, and torque.	feedback
`getEEPose()`, `getToolPose()`	Read current end-effector/tool cartesian pose.	feedback
`isArmJointEnabled()`	Read per-joint arm enable state.	status
`getStatus()`, `printStatus(status)`	Read or print aggregated runtime status (`SystemStatus`).	status
`getConfigurator()`, `getKinematics()`	Access specialized subsystems for config/kinematics.	subsystems

Configuration API (`Configurator`)

API	Description
`setToolOffset(offset), getToolOffset()`	Set or get tool frame offset from end-effector frame.
`setArmPosLimits(limit), setArmVelLimits(limit)`	Set joint position/velocity safety limits.
`getArmPosLimits(), getArmVelLimits()`	Get active joint safety limits.
`getArmMaxPos(), getArmMaxVel()`	Get hardware max position/velocity constraints.
`getArmMaxPosJump(), getArmMaxVelJump()`	Get anti-jump protection limits.
`getArmMaxPosFollowError(), getArmMaxVelFollowError(), getArmMaxTorFollowError()`	Get following-error thresholds.
`setGripperPosLimits(limit), getGripperPosLimits()`	Set or get gripper position limits.
`getGripperMaxPos()`	Get gripper hardware max range.

Kinematics API (`Kinematics`)

API	Description
`forwardKinematics(arm_pos)`	Compute cartesian pose from 6-joint arm state.
`inverseKinematics(ee_pose, arm_pos)`	Solve 6-joint arm state for a target cartesian pose; returns `bool`.
`eulerToQuaternion(euler)`	Convert intrinsic XYZ Euler angles to quaternion.
`quaternionToEuler(quat)`	Convert quaternion to intrinsic XYZ Euler angles.
`eeToToolPose(ee_pose, tool_offset)`	Transform end-effector pose into tool pose.
`toolToEEPose(tool_pose, tool_offset)`	Transform tool pose into end-effector pose.

Types (`types.h`)

Type / Enum	Description
`Position`, `Quaternion`, `EulerAngle`	Cartesian position, orientation, and intrinsic XYZ Euler angles.
`Pose`	Cartesian pose = `Position` + `Quaternion` orientation.
`JointState6f / JointState1f`	6-DOF arm joint values / 1-DOF gripper value.
`JointState6b / JointState1b`	6-DOF arm enable flags / gripper enable flag.
`JointState6u / JointState1u`	Per-joint diagnostic bitmasks for arm and gripper.
`Path<T>`	Waypoint sequence (`std::vector<T>`).

Type / Enum	Description
`RobotJointStatef` / `RobotJointStateb`	Combined arm+gripper state in float or bool forms.
`JointLimits<T>`, `JointLimits6f` / `JointLimits1f`	Per-joint min/max limits for arm and gripper.
`SmoothingMethod`	`kLinear`, `kCos`, `kCubic`, `kQuintic`, `kNone`.
`RobotState`	`kStartup`, `kIdle`, `kMoving`, `kSettling`, `kError`, `kRecovery`, `kShutdown`.
`PlanResult`	`kSuccess`, `kPoseNotReachable`, `kLinearPathFailed`.
`SystemStatus`	Aggregated status: `robot_state`, `plan_result`, `robot_diagnostic_flags`, per-joint diagnostic flags.
`DiagnosticFlags`	Bitmask flags for target/actuator saturation, jump limits, boundary safety, planner events, and hardware/tracking faults.